

SLOPEWATCH · DESIGN DECISION RECORD

SlopeWatch recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SLW-DDR-002	Version	0.2
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/slopewatch		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-02	Amish Chadha	01 to 03 decided by Amish on 2026-10-02 (recommendations approved, SLW-DEC-001)
0.1	2026-09-25	Amish Chadha	Record the recommendations accepted by Amish on 2026-09-25, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item below with a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items without a recommendation (O1 to O3) were decided on 2026-10-02 (SLW-DEC-001).

■ Context

On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." This record lists every SlopeWatch item that carried a recommendation in `docs/REVIEW.md` (TRL 2 and TRL 3 sessions) or in SLW-DDR-001, and what changed in the repo because of it. Where a recommendation offered several options, the recommended option is the decision. Work that would need TRL 4 (build, test, trial, firmware beyond a sketch, purchasing) is decided but on hold, because TRL 4 is on hold by Amish's instruction. Changes that belong in another repo are listed as cross-repo actions and were not made here.

■ Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

#	Item	Decision	What changed in this repo
1	SLW-DDR-001 D1, budget (TRL 2 item 1)	Option B: FieldNode core costed in the FieldNode project; \$250 covers SlopeWatch-specific parts	Status wording in SLW-DDR-001 v0.2, SLW-PRB-001 v0.4, SLW-PRC-001 v0.4 and <code>bom/bom-notes.md</code> . <code>budget_usd</code> stays \$250 (no new figure was recommended)
2	SLW-DDR-001 D2, tailings dams in the pitch (item 2)	Keep them, with the limits stated	Status wording only; pitch unchanged
3	SLW-DDR-001 D3, sensing method (item 3)	Surface tilt stakes plus one crack gauge	Status wording only
4	SLW-DDR-001 D4, bus (item 4)	Wired RS-485 bus to one FieldNode; wireless stakes later	Status wording only
5	SLW-DDR-001 D5, inclinometer (item 5)	Murata SCL3300 in mode 1, ADXL355-class fallback	Status wording only
6	SLW-DDR-001 D6, alert rules (item 6)	Published 0.01 and 0.1 degrees per hour thresholds as defaults; two-reading confirmation; keyed 30 min silence that re-arms	Status wording only
7	SLW-DDR-001 D7, alert unit location (item 7)	On the pole or mast at the toe; second unit optional after co-design	Status wording only
8	TRL 3 review item 4, R11 gap with a new mast	Option (a): the existing-pole case is the reference site; the mast is a site option	R11 in SLW-REQ-001 v0.4 restated for the reference site on an existing pole; BOM line 8 marked as a site option; SLW-CAL-001 v0.2 section K recomputed. R11 moves from not met (\$261) to met on paper (\$245, with item 10 below). <code>budget_usd</code> unchanged
9	TRL 3 review item 5, FieldNode 12 V rail (R8)	Option (a): ask FieldNode to specify at least 0.5 A continuous on its 12 V rail	Recorded as a cross-repo action in <code>docs/REVIEW.md</code> ; R8 text updated. FieldNode not edited; R8 stays at risk until FieldNode rates the rail
10	TRL 3 review item 6, keyed switch location (safety, R7)	Move the switch about 5 m from the mast on its own lead	Design change: <code>cad/src/model.py</code> adds <code>switch_post()</code> (26.9 mm post, 1.4 m above ground, 0.5 m driven, switch box at 1.3 m, 7 m lead) and removes the box from the mast; new <code>cad/step/slopewatch-switch-post.step</code> and <code>.stl</code> ; SLW-DWG-001 Rev P1 to P2; BOM line 7 \$25 to \$33; media re-rendered; SLW-CAL-001 v0.2 adds [E5]: level at the switch 105.4 to 95.5 dB(A), NIOSH allowance 4.3 to 43 min; mast base moment 550 to 541 N·m, factor 2.0 to 2.1
11	TRL 3 review item 7,		Status wording in SLW-PRC-001 v0.4 and SLW-REQ-001 v0.4; no number changed

#	Item	Decision	What changed in this repo
	precaution beacon duty	Keep the slow flash at 1 % duty or less (as set by SLW-CAL-001)	
12	TRL 3 review item 8, networks at SF12	A site that needs SF12 uses its own TwinKit gateway rather than The Things Network	Network rule added to SLW-REQ-001 v0.4 (R4 and assumptions), SLW-PRC-001 v0.4 and SLW-CAL-001 v0.2 section D. The gateway stays outside the site cost

■ Items left open on 2026-09-25

Table 2. Items with no recommendation here, decided on 2026-10-02.

#	Item	Status
01	The <code>problem</code> wording in <code>project.yaml</code> ("often give warning"), changed at TRL 2 with "accept or revert" and no recommendation. The wording stays in place	Decided by Amish, 2026-10-02 (recommendation approved, SLW-DEC-001): the <code>problem</code> wording "often give warning" is accepted
02	First co-design partner and site type; no preference stated	Decided by Amish, 2026-10-02 (recommendation approved, SLW-DEC-001): first partner and site type to approach is an operating aggregate quarry with a geotechnical engineer on staff, testing on a bench away from the work face; a hillside community and its district disaster office only after TRL 4
03	Crack-gauge thresholds (placeholders 1 mm per day over 24 h and 1 mm per hour), to be set with a geotechnical partner; no recommendation	Decided by Amish, 2026-10-02 (recommendation approved, SLW-DEC-001): a geotechnical partner sets thresholds per site before any alarm goes live; until then the published tilt-rate thresholds and the crack-gauge placeholders are for logging only, not for public alarms

■ Consequences

- Requirement status (SLW-CAL-001 v0.2): 0 not met (was 1), 4 at risk (R6, R7, R8, R9), 5 met on paper (was 4), 2 met by design, 2 not verifiable at TRL 3.
- Cost: \$245 SlopeWatch-specific parts for the reference site against \$250 (was \$261 with a new mast, \$237 on a pole without the switch post); \$269 at a site that needs the mast; \$371 for the reference site with the FieldNode core.
- Cross-repo actions, not made here: FieldNode to rate its 12 V rail at 0.5 A or more continuous (item 9); FieldNode's proposed airtime rule (longer intervals at SF10 and slower on The Things Network) needs an exception for SlopeWatch alert states, or such sites use a private gateway, which item 12 now requires at SF12.
- On hold (TRL 4): the suggested cut-cable and flat-battery fault message as a firmware rule, the capsule chamber test in CalRig and every build, test or field trial. TRL 4 is on hold by Amish's instruction.
- `trl` and `trl_target` stay at 3.